

Which Arab? How LLMs Treat Candidates from the 22 Arab League Countries in Hiring and Promotion

Project proposal, IS 617 (HWS26)

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This is the plan for our course project. We would like your feedback on it before the pitch on 13 October. Our questions are in Section 6.

1 Research questions

We give an LLM a job and CVs that are identical except for one line stating the nationality, and observe what it decides. The 22 Arab League countries differ widely in wealth, in political stability and in how their citizens are seen abroad: high-income Gulf states, middle-income countries such as Egypt, Jordan and Morocco, and countries marked by war and displacement such as Syria, Yemen, Sudan, Libya and Somalia. We ask whether a model carries these differences into decisions about individual people.

- RQ1** Do fit scores and decisions differ across the 22 Arab League nationalities when everything else is identical, and is this spread larger than for unrelated placebo nationalities? This is the main question.
- RQ2** Do the differences follow a country's economic and political position, for example Gulf nationalities above nationalities associated with war, displacement or low income?
- RQ3** Does the place of the job matter? Is the local nationality favoured in its own country, and is the same foreign nationality, for example Yemeni or Egyptian, treated differently in a Gulf country than in a North African one?
- RQ4** Do the differences remain when the model is told that nationality is not a criterion, and when it may let a lottery decide between two identical candidates?
- RQ5** Are the patterns the same in hiring and in promotion, and across models?

The comparison between Arab and non-Arab nationalities is secondary.

2 Why the question is realistic and open

The scenario is common. Many Arabs work in an Arab country other than their own. In the Gulf states, Arabs from other countries were 72% of the foreign population in 1975 and still about a third in the early 2000s (Kapiszewski, 2006). In Jordan, Jordanian, Egyptian and Syrian workers share one labour market (Razzaz, 2017). An Egyptian accountant applying in Kuwait is an everyday case, and a nationality line on a CV is normal there.

Nationality is used openly. A report for the International Labour Organization finds that segmentation in Jordan is based on nationality rather than on education or skills (Razzaz, 2017). Among low-income workers in Qatar, pay differs by nationality (Gardner et al., 2013). Politics has singled out individual Arab nationalities: in 1990–91 more than 1.5 million people were expelled from Gulf states, among them up to one million Yemenis from Saudi Arabia, 200,000 Jordanians and 150,000 Palestinians (Kapiszewski, 2006).

Models hold country-specific associations. In text generated by GPT-2, Libyan, Sudanese and Tunisian were among the five demonyms with the most negative sentiment (Narayanan Venkit et al., 2023). In simulated immigration decisions, GPT models suggested additional scrutiny for applicants from Iraq, Sudan and Somalia (Mao and Zhao, 2025). Models also represent the values of some Arab countries much better than those of others (Zahraei et al., 2025).

The answer is open in both directions. Field experiments in Western labour markets find about 41% fewer positive callbacks for applicants with Arab, Maghrebi or Middle Eastern names (Lipens et al., 2023), and origin can matter within that group: Swedish employers called back 10% of Iraq-born and 5% of Somalia-born applicants (Vernby and Dancygier, 2019). Inside the region the evidence points the other way. Qatari citizens made

no significant distinction between nationalities in a survey experiment on residency (Shockley and Gengler, 2024), and Arab expatriates in Kuwait and Qatar favour “fellow Arabs” as one group (Blaydes and Gengler, 2023). A result of “no difference” would therefore also be an answer.

Existing LLM audits cannot tell. Audits of LLMs in hiring mostly vary race and gender through names (Gaebler et al., 2024; Armstrong et al., 2024). Where Arab applicants appear, they are one group: a penalty of 1.41 points on a 100-point scale for Arab names in GPT-3.5 (Lippens, 2024), and an average effect close to zero in a more recent study (Hoffmann et al., 2026). This follows from the method, because a name does not identify a country: raters assigned only 34% of Moroccan names to Morocco (Martiniello and Verhaeghe, 2022). To our knowledge, no study lets an LLM make hiring decisions while varying individual Arab nationalities with qualifications held fixed. The closest one we found pools them into Arab versus non-Arab (Albaroudi et al., 2026). We treat this claim as tentative.

3 Design

Materials. Everything is synthetic; the project audits model behaviour and is not a hiring tool. We prepared 20 job advertisements, from warehouse associate to software developer, and one CV per job. Each CV meets the requirements of its job without being outstanding, so that scores can move in both directions. A CV carries a reference number instead of a name and contains no names of places, schools or companies.

Manipulation. One line in the CV, for example Nationality: Jordanian. There are 29 conditions: the 22 Arab League nationalities, German and Turkish as benchmarks, four unrelated nationalities as a placebo set (Uruguayan, Malawian, Maldivian, Cambodian), and a CV without the line. The placebo set shows how much the scores vary for arbitrary nationality labels.

Setting. The model is told that it recruits for the company in a given country. Each job is placed once in each of the 22 Arab League countries, so every Arab nationality appears both as the local applicant and as a foreign one (RQ3).

Tasks. (1) Rate one applicant: a fit score from 0 to 100 and interview yes or no. (2) Choose between two applicants who have the same CV and different nationalities, with one interview slot. In one

Block	Runs per model
1 Hiring, rate one applicant	12,760
2 Hiring, choose between two	19,360
3 Blocks 1 and 2 with the added paragraph	32,120
4 Promotion, all of the above	64,240
Total	128,480

Table 1: Planned runs per model, in order of priority.

version the model may answer A, B or “lottery”; in the other it must choose A or B. Each pair is shown in both orders. (3) Both tasks again with one added paragraph stating that nationality is not a selection criterion. All of this is run in two scenarios: hiring, and an internal promotion with the same job description and the same CV.

Models. Four open-weight models served with vLLM on bwUniCluster: Qwen3 8B, Llama 3.1 8B, Gemma 3 12B and Mistral Small 3.2 24B.

Size. Table 1 lists the runs per model. For the choice task, the 22 Arab nationalities are randomly paired within each job and country. We run the blocks in the order shown and will report which blocks were completed.

4 What we will compare

RQ1 How far the mean score and the decision rates differ across the 22 Arab nationalities, against the spread across the placebo nationalities.

RQ2 Each nationality’s result against its country’s income group (World Bank) and against whether the country is affected by armed conflict. The conflict classification still has to be fixed.

RQ3 Local versus foreign applicants, and each foreign nationality across the countries of the job.

RQ4 The same comparisons with the added paragraph; in the choice task, which nationality is chosen, how often the model takes the lottery, and what it does when the lottery is removed.

RQ5 Hiring against promotion, and the four models against each other.

We have not fixed the statistical analysis yet. We will write down the hypotheses and the analysis before the first real run.

5 Timeline

Pitch on 13 October. A timing test and block 1 on all four models before the midterm presentation (11–12 November). The remaining blocks before the final presentation (9–10 December). Report due on 8 January 2027.

6 Questions

1. Is the scope right for a short paper, or should we drop the promotion scenario and keep hiring only?
2. For RQ2, which classification of the countries would you use (income, conflict, region), and is it defensible to group them at all?
3. Is one CV per job at one qualification level enough, or would you expect several levels?
4. Is the choice between two identical CVs with a lottery option a sensible measure, and are four placebo nationalities enough as a yardstick?
5. Are four open-weight models between 8B and 24B parameters acceptable?
6. Do you see concerns with reporting results by individual nationality?

AI use. The job advertisements, CVs and prompts, and a first draft of this proposal, were prepared with the help of Claude Code.

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